

A
REPORT ON
SUMMER INTERNSHIP

at
AICW, Edunet Foundation
submitted by

Dhruti Vipulbhai Soni (230490131112)

in fulfilment for
B.E. SEMESTER–VII
in
DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING



of

S. N. PATEL INSTITUTE OF TECHNOLOGY,
BARDOLI, SURAT

GUJARAT TECHNOLOGICAL UNIVERSITY, AHMEDABAD

Academic Year

2026-2027



S. N. Patel Institute of Technology, Bardoli

Department of Computer Science & Engineering

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CERTIFICATE

This is to certify that the report submitted has been carried out by **Dhruti Vipulbhai Soni (230490131112)** under my guidance in fulfillment for the course named Summer Internship (3170001) in the degree of **Bachelor of Engineering** in Department of **Computer Science & Engineering** at Sitarambhai Naranji Patel Institute of Technology, Bardoli, Surat of Gujarat Technological University, Ahmedabad during the academic year 2025-26. He has done his Summer Internship in **AICW , Edunet Foundation** and successfully completed all activities related to Summer Internship under my guidance.

Internal Guide

Prof. Hemangini A. Patel

I/C H.O.D

Mr. Bhavesh D. Patel

ACKNOWLEDGEMENT

Internship is an indispensable part of any engineering curriculum. It provides the students an opportunity to gain experience about the practical application of their knowledge.

My internship at **AI Careers For Women , Edunet Foundation** provides me experience in not only the technical and practical accepted of the industry but also provide the great insights into the actual working of the industry.

I would like to express my sincere gratitude to **AICW, Edunet Foundation** for providing me with the opportunity to undergo my Summer Internship and gain valuable practical exposure in the field of Computer Science & Engineering.

I am deeply thankful to our External Guide **Mr. Sharat Rokkam** and Internal Guide, **Prof. Hemangini A. Patel**, for his constant guidance, encouragement, and valuable suggestions throughout the duration of this internship, which helped me complete this report successfully.

I would also like to extend my heartfelt thanks to **Mr. Bhavesh D. Patel**, I/C H.O.D of the Department of Computer Science & Engineering, for his support and for providing the necessary facilities to carry out this internship.

Finally, I would like to thank **S. N. Patel Institute of Technology, Bardoli** and **Gujarat Technological University** for giving me this platform to enhance my skills and knowledge.

Dhruti Vipulbhai Soni

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Chapter 1: Introduction of Industry

1.1 About the Organization

Edunet Foundation is a non-profit organization working in the areas of skilling, livelihood enhancement, and STEM education.

Founded in 2015, Edunet Foundation is registered under sections 12AA and 80G of the Income Tax Act.

The Foundation has been a trusted implementation partner for top global technology companies and has been certified as a Great Place to Work for five consecutive years.

It also holds Special Consultative Status with the United Nations Economic and Social Council. My internship was carried out under **AI Careers for Women (AICW)**, a national initiative led by Edunet Foundation.

AICW is a national initiative by Edunet Foundation that empowers Non-STEM women with future-ready AI skills, mentorship, and certification, delivered in collaboration with Microsoft, SAP and LinkedIn, with support from the Ministry of Skill Development & Entrepreneurship.

1.2 Mission and Vision

AICW was born from a critical need: while women make up nearly half of India's graduates, they comprise less than 30% of its tech workforce. The program directly addresses this gap, with a mission to build an ecosystem enabling over 100,000 women to pursue tech careers within five years, using structured training, certifications, and mentorship.

1.3 Business/Program Activities

The AICW program uses a practical, multi-layered learning approach that integrates instructor-led sessions for foundational knowledge, self-paced activities for deeper understanding, and capstone projects that reinforce learning through real-world application. My track within the program focused on Generative AI, covering the underlying data science and machine learning foundations that support generative AI systems. Edunet Foundation runs several such programs beyond AICW, spanning school, higher-education, and vocational learners.

1.4 My Role in the Organization

During my internship under the AI Careers for Women (AICW) program by Edunet Foundation, I was primarily engaged in data preparation and machine learning tasks that fed into a Generative AI pipeline, using Python. My key responsibilities included cleaning and preprocessing datasets to handle missing values, inconsistencies, and outliers, ensuring the data was accurate and structured for downstream use. I worked extensively with Python libraries such as Pandas and NumPy for data manipulation and numerical computation, and used Scikit-learn to build and evaluate machine learning models that supported the broader generative AI workflow. This hands-on exposure helped me understand how traditional data science and ML practices integrate into generative AI systems, and strengthened my practical skills relevant to a career in AI and Generative AI.

Chapter 2: Details of the Internship

2.1 Internship Position

During the GTU Summer Internship 2026, I was selected for the AI/ML domain with the organization AI Careers for Women (AICW) program, conducted by Edunet Foundation.

Official designation: AI Intern

Duration: 4 weeks (06-07-2026 to 31-07-2026)

Mode: Offline

The internship involved a bunch of self learning, expert guidance & work implementation using Python & its Libraries, VS Code IDE, Datasets.

2.2 Assigned Tasks and Daily Activities

My mentor / guide had assigned tasks and activities on a weekly and daily basis, which included understanding the project requirements, dataset creation and preprocessing, performance analysis, learning progress tracking, AI-based insight generation, data visualization, dashboard development, testing, and documentation work for the **AI Student Performance & Learning Progress Tracker**.

Week 1 – Data Management & Preprocessing

a) Documentation go-through:

- Understanding the project requirements and workflow.
- Understanding student performance and learning progress parameters.
- Understanding CSV dataset structure and required attributes.
- Studying Pandas for data manipulation and preprocessing.

b) Setup & Installations:

- Python environment setup.
- Installation of Pandas and NumPy.
- Setting up the project folder structure.
- Creating and organizing the data directory.

c) Learning Based Tasks:

- Introduction to Python for data analysis.
- Reading and writing CSV files using Pandas.
- Creating DataFrames.
- Selecting, filtering, and modifying data.

d) Project Work:

- Creation of **students.csv**.
- Collection and organization of student academic and learning data.
- Implementation of **data_preprocessing.py**.
- Cleaning and validating the student dataset.

Week 2 – Performance Analysis**a) Documentation go-through:**

- Understanding student academic performance metrics.
- Understanding grading and ranking systems.
- Studying methods for statistical analysis using Pandas and NumPy.

b) Learning Based Tasks:

- Data aggregation using Pandas.
- Calculating totals and averages.
- Working with conditional statements for grade calculation.
- Understanding percentage and pass/fail criteria.

c) Project Work:

- Implementation of **performance.py**.
- Calculation of total marks for each student.
- Calculation of percentage.
- Assignment of grades based on percentage.
- Determination of Pass/Fail status.
- Calculation of student ranks.
- Identification of overall topper.
- Identification of subject-wise toppers.
- Calculation of average marks.
- Identification of highest and lowest marks.
- Verification of calculated performance results.

Week 3 – Learning Progress & AI Insights

a) Documentation go-through:

- Understanding learning progress indicators.
- Studying the relationship between attendance, study hours, quiz scores, and assignment completion.
- Understanding rule-based AI recommendation systems.

b) Learning Based Tasks:

- Data filtering and conditional analysis using Pandas.
- Implementing if-else based decision logic.
- Working with attendance percentages.
- Analyzing quiz scores.
- Tracking study hours.
- Tracking assignment completion.
- Designing rule-based recommendation conditions.
- Generating personalized student learning suggestions.

c) Project Work:

- Implementation of **progress.py**.
- Tracking student attendance.
- Tracking quiz scores.
- Tracking study hours.
- Tracking assignment completion.
- Implementation of **ai_insights.py**.
- Generating AI-style recommendations based on predefined rules.
- Implementing recommendations such as:
 - Attendance below 75% → Improve attendance.
 - Quiz score below 50 → Revise previous topics.
 - Study hours below 2 hours → Increase study time.
 - Percentage above 90 → Excellent work.
- Testing recommendations against different student performance conditions.

Week 4 – Visualization, Dashboard & Final Integration

a) Documentation go-through:

- Understanding data visualization techniques.
- Studying different chart types and their applications.
- Understanding dashboard design and report presentation.
- Reviewing the complete project workflow.

b) Setup & Installations:

- Installation and configuration of Matplotlib.
- Installation and configuration of Seaborn.
- Preparing the environment for data visualization.

c) Learning Based Tasks:

- Creating bar charts using Matplotlib.
- Creating line charts for progress analysis.
- Creating pie charts for categorical data.
- Creating histograms for marks distribution.
- Creating heatmaps for subject-wise performance analysis.
- Understanding chart labels, titles, legends, and axes.
- Combining multiple visualizations into a dashboard/report.

d) Project Work:

- Implementation of **visualization.py**.
- Creation of student performance bar charts.
- Creation of learning progress line charts.
- Implementation of **dashboard.py**
- Integration of preprocessing, performance analysis, progress tracking, and AI insights.
- Combining analytical results and visualizations into a single dashboard/report.
- Testing the complete project workflow.
- Debugging and correcting errors.
- Final documentation and project report preparation.

2.3 Tools and Technologies Used

Category	Tools / Technologies
Programming Language	Python
Data Handling	Pandas
Numerical Analysis	NumPy
Data Visualization	Matplotlib, Seaborn
Dataset Format	CSV
Development Environment	VS Code
Dashboard & Reporting	Matplotlib, Seaborn, Python
Operating System	Windows

2.4 Skills Learned

During the internship, I developed a range of technical skills including data cleaning and preprocessing techniques to handle missing values, duplicates, and outliers in real-world datasets. I gained hands-on experience with Python libraries such as Pandas and NumPy for data manipulation, and Matplotlib and Seaborn for exploratory data visualization.

I also learned to build and evaluate machine learning models using Scikit-learn, and understood how these foundational data science practices feed into larger Generative AI pipelines. Beyond technical skills, I learned to document my workflow systematically and communicate progress clearly during daily mentor check-ins.

2.5 Project Completed

Problem Statement: Educational institutions often maintain student information such as marks, attendance, quiz scores, assignments, and study hours in separate systems or spreadsheets. This makes it difficult for teachers to analyze a student's overall academic performance and learning progress efficiently. Manual analysis is time-consuming, prone to errors, and does not provide personalized feedback. As a result, identifying weak students, tracking improvement over time, and offering timely suggestions becomes challenging.

Solution: The AI Student Performance & Learning Progress Tracker is a Python-based analytics system that automatically analyzes student data from a CSV file. It calculates total marks, percentage, grades, pass/fail status, and overall performance. The system also tracks learning progress by evaluating attendance, quiz scores, assignment completion, and study hours. Using rule-based AI, it generates personalized suggestions to help students improve in weaker areas. Additionally, interactive visualizations and a dashboard provide teachers and students with clear insights into academic performance, making the evaluation process faster, smarter, and more effective.

Part A: Data Management & Preprocessing — Created and cleaned the student CSV dataset, handled missing and duplicate records, validated data types, and prepared the dataset for analysis.

Part B: Performance & Progress Analysis — Calculated total marks, percentage, grades, pass/fail status, ranks, toppers, averages, attendance, quiz scores, study hours, and assignment completion.

Part C: AI Insights & Recommendations — Developed rule-based AI suggestions based on student performance and learning progress, such as improving attendance, increasing study hours, and revising weak topics.

Integration — Combined data preprocessing, performance analysis, learning progress tracking, AI recommendations, and visualizations into a single student performance tracking dashboard/report. The project brought together Python, Pandas, NumPy, Matplotlib, and Seaborn skills into a practical academic analytics solution.

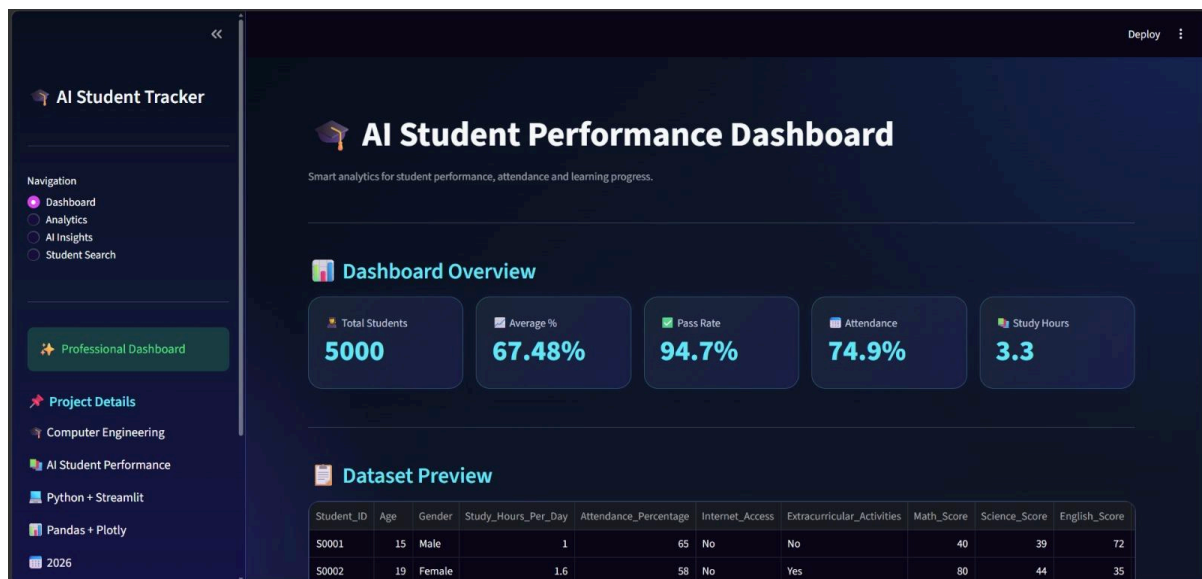


Figure 1: Student Dashboard

Explanation: Student Dashboard can give you information about enrolled students number, passing rate, available datasets, AI insights, project details.



Figure 2: Analytics Section

Explanation: The analytics section can give you the visual presentation of information like average score, score in a particular subject, comparison of study hours & final percentage with the help of graphs.

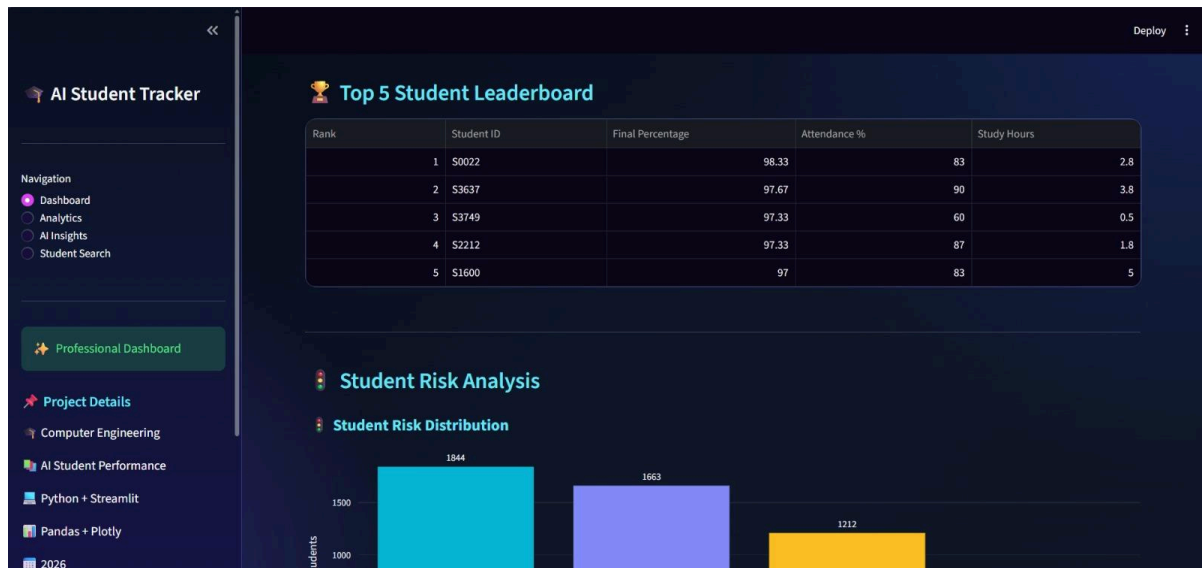


Figure 3: Leaderboard

Explanation: The leader board displays the name & other information of the top performing students.

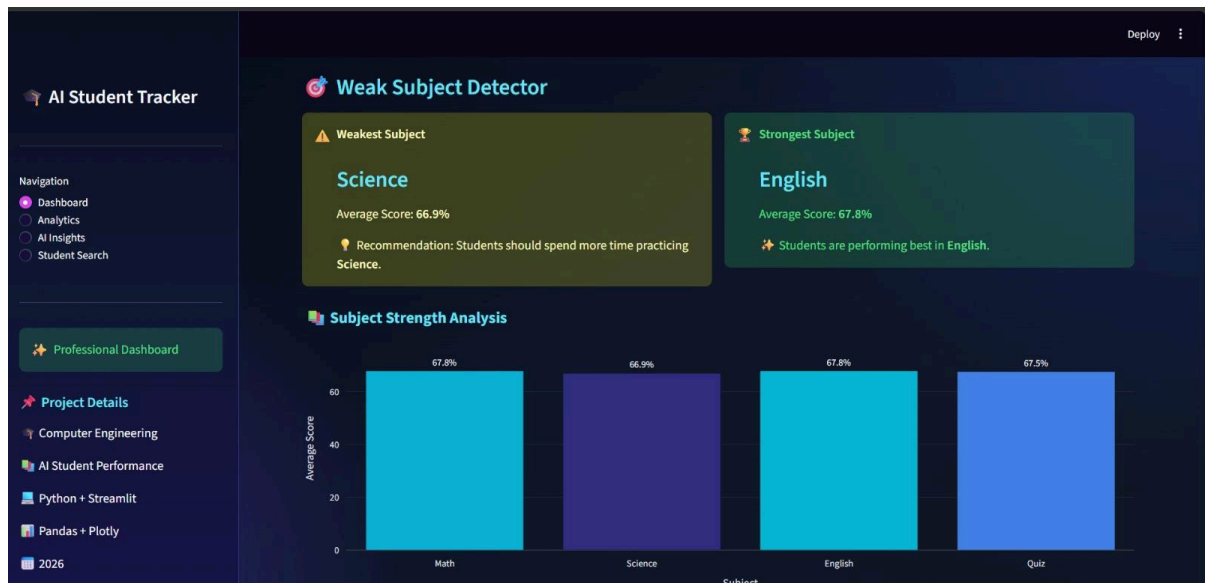


Figure 4: Weak Subjects Detector

Explanation: This section can help you find out the weak subjects & provide suggestions to improve in that particular subject.

Chapter 3: Skills Developed During Internship

3.1 Technical Skills Acquired

During my internship, I gained practical experience in Python-based data analysis and visualization through the development of the AI Student Performance & Learning Progress Tracker. Some of the key technical skills I developed include:

- **Python & Data Handling:** Used Python and Pandas to read, clean, transform, and manage student datasets, including handling missing values, duplicate records, and data-type inconsistencies.
- **Data Analysis:** Used Pandas and NumPy to calculate total marks, percentages, grades, ranks, averages, and other student performance metrics.
- **Data Visualization:** Used Matplotlib and Seaborn to create bar charts, line charts, pie charts, histograms, and heatmaps for analyzing and presenting student performance.
- **Dashboard Development:** Combined analytical results and visualizations into a consolidated dashboard/report for easier interpretation of student performance.
- **CSV & File Handling:** Gained practical experience in creating, reading, processing, validating, and saving CSV datasets using Python.

3.2 Soft Skills & Professional Development

Apart from technical skills, the internship also helped me improve various soft skills:

- **Communication Skills:** Improved the ability to explain technical progress, problems, and solutions clearly during discussions with the mentor and team members.
- **Team Collaboration:** Worked with team members by dividing project responsibilities and integrating individual contributions into one complete application.
- **Time Management:** Worked within a structured four-week timeline, completing data preprocessing, performance analysis, progress tracking, AI insights, and dashboard development in planned phases.
- **Problem Solving:** Identified data quality issues, debugging errors, and visualization problems and resolved them through systematic testing and analysis.

3.3 Tasks Performed During Internship

During the internship, I worked on a series of tasks that helped me develop practical skills in Python, Data Analysis, and Visualization. The major tasks included:

- Creating and organizing the student CSV dataset.
- Cleaning and preprocessing student data using Python and Pandas.
- Handling missing values, duplicates, and inconsistent data types.
- Calculating student performance metrics such as percentage, grade, rank, and Pass/Fail status.
- Performing subject-wise and overall performance analysis.
- Tracking attendance, quiz scores, study hours, and assignment completion.
- Developing rule-based AI recommendations for improving student performance.
- Creating different visualizations using Matplotlib and Seaborn.
- Developing and integrating a student performance dashboard/report.
- Testing the complete workflow and preparing project documentation.

3.4 Connection to Academic Learning

This internship connected directly to concepts from my academic coursework:

- **Programming Fundamentals:** Python programming concepts learned during academic studies were applied to develop data processing, analysis, and visualization modules.
- **Data Analysis:** Statistical and analytical concepts were practically applied to evaluate student performance and identify meaningful patterns.
- **Data Visualization:** Academic knowledge of data visualization was applied to create charts and graphical representations using Matplotlib and Seaborn.
- **Database & Data Management Concepts:** Concepts related to structured data and data management were applied while designing, cleaning, validating, and maintaining the student dataset.
- **Artificial Intelligence:** Basic AI concepts were applied through rule-based logic to generate personalized learning recommendations.

3.5 Career Impact & Future Relevance

This internship contributed significantly to my technical and professional development:

- **Career Confidence:** Gained practical confidence in Python, data analysis, visualization, and basic AI implementation.
- **Portfolio Project:** The **AI Student Performance & Learning Progress Tracker** provides a practical project that can be demonstrated in my resume and technical interviews.
- **Learning Foundation:** Developed a strong foundation in data preprocessing, analysis, visualization, and rule-based AI systems.
- **Professional Growth:** Gained experience working in a structured project environment, collaborating with team members, following a development workflow, and documenting technical work.
- **Future Relevance:** The skills developed during the internship provide a foundation for further learning in **Data Science, Machine Learning, Artificial Intelligence, and Educational Analytics**.

3.6 System Architecture

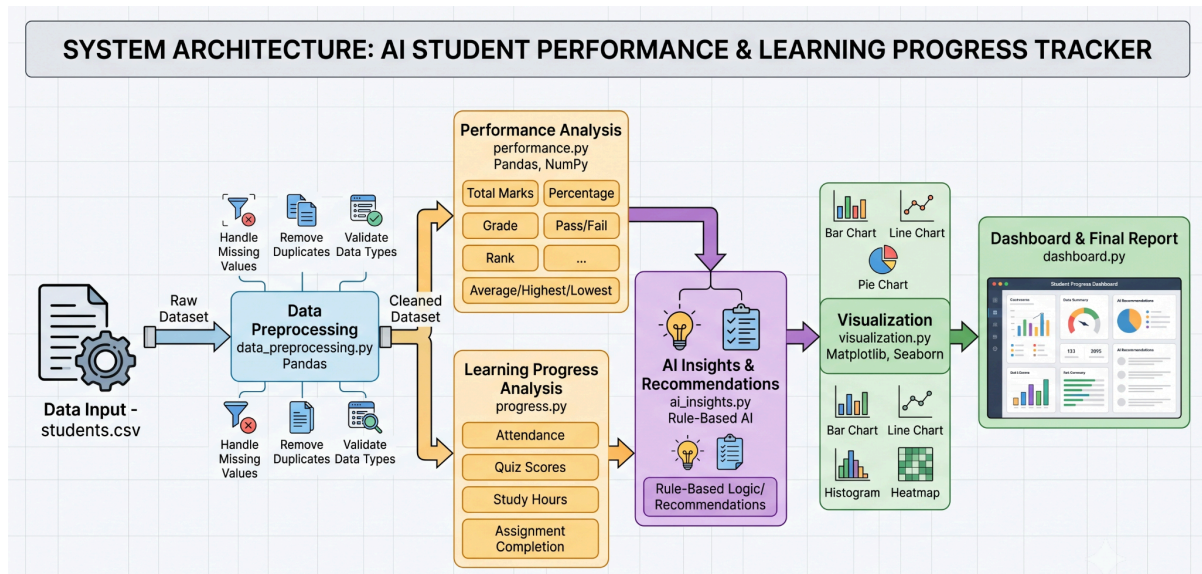


Figure 5: System Architecture Diagram

The system architecture of the **AI Student Performance & Learning Progress Tracker** represents the complete flow of student data from input to final analysis and reporting.

The process begins with the **students.csv** dataset, which contains student details such as marks, attendance, quiz scores, study hours, and assignment completion. The data is first processed using **data_processing.py**, where missing values and duplicate records are handled and data types are validated.

The cleaned data is then used by **performance.py** to calculate total marks, percentage, grade, pass/fail status, rank, average marks, and topper details. At the same time, **progress.py** analyzes attendance, quiz scores, study hours, and assignment completion.

The results are passed to **ai_insights.py**, which uses rule-based conditions to generate learning recommendations for students. After this, **visualization.py** creates different charts such as bar charts, line charts, pie charts, histograms, and heatmaps.

Finally, **dashboard.py** combines the analysis, recommendations, and visualizations into a consolidated **Student Performance and Learning Progress Dashboard/Report**, making the results easier to understand and interpret.

Chapter 4: Conclusion

The AI Student Performance & Learning Progress Tracker was developed to analyze students' academic performance and learning activities in a structured manner. The project involved data collection, preprocessing, performance analysis, progress tracking, rule-based AI recommendations, and data visualization using Python, Pandas, NumPy, Matplotlib, and Seaborn. The system helps identify students' strengths and areas requiring improvement through performance metrics and personalized suggestions. The visualizations and dashboard provide a clear representation of the analyzed data, making it easier to understand overall student progress. Overall, the project provided practical experience in data preprocessing, data analysis, visualization, and basic AI implementation, while demonstrating how data-driven techniques can be applied to support better monitoring of student performance and learning progress.